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Quang Dao

Education

2022–Present **Carnegie Mellon University**, *Pittsburgh, PA*
PhD in Computer Science. *Advisors:* Aayush Jain and Riad Wahby.

2020–2022 **University of Michigan**, *Ann Arbor, MI*
MA in Mathematics. *Advisor:* Paul Grubbs

2016–2020 **Columbia University**, *New York, NY*
BA in Mathematics and Computer Science

Research Interests

My research focuses on the **design**, **formal verification**, and **performance optimization** of **succinct zero-knowledge proofs (SNARKs)**. I founded and co-lead the development of **ArkLib**, a Lean library for compositionally verifying SNARK constructions, and am also a co-lead developer of **VCV-io**, a Lean framework for reasoning about cryptographic protocols that ArkLib builds on. I contribute to **Jolt**, a state-of-the-art zero-knowledge virtual machine (zkVM). My earlier work also explored foundational post-quantum cryptography from code-based assumptions.

Publications

(* = first author(s))

8. Behzad Abdolmaleki, Matteo Campanelli, **Quang Dao**, Hamidreza Khoshakhlagh. On the Simulation-Extractability of Proof-Carrying Data. *PKC 2026*.
7. Carl Kwan*, **Quang Dao***, Justin Thaler. Verifying Jolt zkVM Lookup Semantics. *Financial Cryptography 2025*.
6. **Quang Dao**, Aayush Jain. Lossy Cryptography from Code-Based Assumptions. *CRYPTO 2024*. ***Best Paper from Early Career Researchers.***
5. **Quang Dao**, Aayush Jain, Zhengzhong Jin. Non-Interactive Zero-Knowledge from LPN and MQ. *CRYPTO 2024*.
4. **Quang Dao**, Yuval Ishai, Aayush Jain, Huijia Lin. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN. *CRYPTO 2023*.
3. **Quang Dao***, Jim Miller*, Opal Wright, Paul Grubbs. Weak Fiat-Shamir Attacks on Modern Proof Systems. *IEEE S&P 2023*. ***Distinguished Paper Award.***
2. **Quang Dao**, Paul Grubbs. Spartan and Bulletproofs are simulation-extractable (for free!). *EUROCRYPT 2023*.
1. **Quang Dao**, Julian Wellman, Calvin Yost-Wolff, Sylvester W. Zhang. Rowmotion Orbits of Trapezoid Posets. *The Electronic Journal of Combinatorics*, P2-29, 2022.

Preprints

5. Behzad Abdolmaleki, Matteo Campanelli, **Quang Dao**, Shadman Mohammadi, Nahid Roustaeifar. Resettable Non-Interactive Zero-Knowledge: Attacks and Defenses. *ePrint 2026*.
4. Devon Tuma*, **Quang Dao***, James Waters, Alexander Hicks, Nicholas Hopper. VCVio: Verified Cryptography in Lean via Oracle Effects and Handlers. *ePrint 2026*.
3. **Quang Dao***, Ari Biswas, Liam Eagen, Andrew Milson, Shahar Papini, Justin Thaler. The Sum-Check Protocol over the Monomial Basis, and Other Optimizations. *ePrint 2026*.
2. **Quang Dao***, Zachary DeStefano*, Suyash Bagad, Yuval Domb, Justin Thaler. Speeding Up Sum-Check Proving (Extended Version). *ePrint 2026. (Combines and improves upon results from ePrint 2025/1117, 2024/1046, and 2024/1210.)*
1. **Quang Dao**, Justin Thaler. Constraint-Packing and the Sum-Check Protocol over Binary Tower Fields. *ePrint 2024*.

Projects

ArkLib github.com/Verified-zkEVM/ArkLib. Lean library for formally verified SNARKs. Founder and co-lead developer.

VCV-io github.com/Verified-zkEVM/VCV-io. Lean framework for reasoning about cryptographic protocols; ArkLib is built on top of it. Co-lead developer.

Jolt github.com/a16z/jolt. State-of-the-art zero-knowledge virtual machine (zkVM) based on structured lookup arguments. Main contributor.

Grants & Fellowships

2025 **Ethereum Foundation Grant**
Grant for the formalization of the FRI-Binius Polynomial Commitment Scheme
Amount: $24,000

2025 **Ethereum Foundation Grant**
Grant for the development of ArkLib, a Lean library for formally verifying SNARKs
Amount: $100,000

2024–2025 **Quad Fellowship**
One-year fellowship for US-based PhD students in STEM fields
Amount: $40,000

2024–2025 **CyLab Fellowship**
One-year fellowship from CyLab at Carnegie Mellon University
Amount: $50,000

Honors & Awards

2024 **Best Paper from Early Career Researchers**
Awarded to the best paper from early career researchers at CRYPTO 2024

2023 **Distinguished Paper Award**
Awarded to top 6% of accepted papers at IEEE Security & Privacy 2023

2020 **Russell C. Mills Award**
Awarded to 2 seniors for excellence in computer science at Columbia

2017 - 2019 **Van Amringe Math Prize**
Awarded annually to the top 3 non-senior students in math at Columbia

2016, 2018 **Putnam Math Competition**. Honorable Mention (top 50)

2016 **International Math Olympiad**. Silver Medal

Industry Experience

Jan 2026 – Present Part-Time Researcher at LayerZero.

Summer 2025 Part-Time Contractor with a16z crypto.

Summer 2024 Research Intern at a16z crypto.

Talks

9. Evolving the Foundations of ArkLib
   - ZKProof 8, Rome (May 2026)
8. ArkLib: Compositional Verification of SNARKs
   - Real World Crypto (Mar 2026)
   - Verified zk(E)VM Seminar (Jan 2026)
   - King's College London (Oct 2025)
   - University of Cambridge (Oct 2025)
   - Ethereum's PQ Interop (Oct 2025)
   - CMU PLunch (Sep 2025)
   - University of Luxembourg Crypto Seminar (July 2025)
   - CRYPTO Day, Telecom Paris (June 2025)
   - EPFL Crypto Seminar (June 2025)
   - CMU Formal Cookies (Oct 2024)
7. Lossy Cryptography from Code-Based Assumptions
   - MIT CIS Seminar (Nov 2024)
   - BUSec Seminar (Nov 2024)
   - NYU Crypto Reading Group (Nov 2024)
   - CMU CyLab Crypto Seminar (Sep 2024)
   - UCLA Crypto Reading Group (Apr 2024)
   - UToronto Theory Seminar (Mar 2024)
6. Non-Interactive Zero-Knowledge from LPN and MQ
   - CMU CyLab Crypto Seminar (Sep 2024)
5. Advanced Security for SNARKs: A Survey
   - a16z Crypto Research Seminar (Jun 2024)
4. Multi-party Homomorphic Secret Sharing and Sublinear MPC from Sparse LPN
   - JP Morgan AlgoCRYPT Seminar (Dec 2023)
   - CMU Crypto Seminar (Nov 2023)
   - NTT Research Seminar (Oct 2023)
   - CyLab Partners Conference (Oct 2023)
   - Vietnam Mathematical Congress (Aug 2023)
3. Weak Fiat-Shamir Attacks on Modern Proof Systems
   - Real World Crypto (Mar 2024)
   - CMU CyLab Security Seminar (Nov 2023)
   - Cornell Security Seminar (Sep 2023)
   - NYU Crypto Reading Group (Sep 2023)
   - Workshop on Attacks in Cryptography (Aug 2023)
   - IEEE S&P (May 2023)

2. Spartan and Bulletproofs are simulation-extractable (for free!)
   - Stanford Crypto Reading Group (May 2023)
   - Telecom Paris Seminar (May 2023)
   - Lattices Meet Hashes Workshop, EPFL (May 2023)
   - CMU Crypto Seminar (April 2023)

Service

Program Committee	ASIACRYPT 2026
External Reviewer	2026: CRYPTO 2025: EUROCRYPT, IEEE S&P, CRYPTO, RANDOM, TCC 2024: STOC, EUROCRYPT, TCC 2023: ASIACRYPT, TCC, FOCS 2022: CRYPTO, EUROCRYPT
Co-Organizer	2022-2025: CMU Crypto Seminar

Teaching

2023-2025	**Western Pennsylvania ARML Team**, *Assistant Coach*, CMU
Fall 2023	**Undergraduate Quantum Computation**, *Teaching Assistant*, CMU
2020–2021	**Calculus I**, *Lead Instructor*, University of Michigan
2017, 2021	**Math & Science Summer Program (MASSP)**, *Mentor*, Vietnam

Mentorship

Chung Nguyen	Post-baccalaureate, 2025–Present
James Waters	Undergraduate, Fall 2025–Summer 2026
Liam Schilling	Undergraduate, Spring & Fall 2025
Ian Martin	Undergraduate, Summer 2025
Jack Zhu	Undergraduate, Spring 2025
Christian Ang	Undergraduate, Spring 2025

Miscellaneous

Languages	English (fluent), Vietnamese (native), French (elementary), Malayalam (elementary)
Other writing	**Quang Dao**, Christina Meng, Julian Wellman, Zixuan Xu, Calvin Yost-Wolff, Teresa Yu. Extended Nestohedra and their Face Numbers. *arXiv 2019*.